

Motor-Skill-Transfer Technology for Piano Playing with Electrical Muscle Stimulation

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ABSTRACT

We introduce a motor-skill-transfer technology using electrical muscle stimulation (EMS) for acquiring piano playing skills. While expert pianists use the coordination of multiple muscles, such as fingers and arms, novices are less aware of muscle coordination and tend to only move their fingers. Our EMS-based system encourages them to use their arms as well as their fingers. Based on the analysis of experts' muscle coordination, our system applies EMS to the novices' forearms and shoulders. With this system, novices should be able to improve their motor skills, such as playing octave tremolos by using wrist rotation to reduce fatigue and playing C major scales more smoothly by coordinating forearm and shoulder muscles to execute the thumb-under technique.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

electrical muscle stimulation, muscle coordination, piano

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1 INTRODUCTION

Electrical muscle stimulation (EMS) has been shown to be effective for acquiring motor skills such as playing musical instruments [Nijima et al. 2022, 2021; Tamaki et al. 2011]. EMS, a key technology for inducing involuntary muscle contraction, has been extensively studied in rehabilitation medicine and has subsequently been applied to assist with motor skill learning. Traditional EMS-based

systems assist novices in learning the kinetic movement of their limbs. However, novices also need to learn to coordinate multiple muscles to achieve optimal performance. To this end, there is a growing need for EMS-based systems that not only guide limb movement but also improve muscle coordination or synergy.

We present an EMS-based assistance system for muscle coordination focusing on two specific piano playing skills, tremolos and scales. Tremolos, which are two notes played rapidly in alternation, and scales, a set of typically seven notes ordered by pitch, are widely used in musical pieces. Novices often struggle with both and tend to use their forearm muscles more [Nijima et al. 2021] and play inconsistently compared to expert pianists [Nijima et al. 2022]. Experts use techniques such as wrist rotation for tremolos [Furuya et al. 2011] and the thumb-under technique to play scales efficiently [Chang 2016], which require the coordinated activation of multiple muscles. Our EMS-based system helps novices by applying EMS to specific muscles involved in these techniques. For tremolos, we apply EMS to the supinator and pronator teres muscles on the forearm involved in wrist rotation. For scales, we stimulate the deltoid muscle on the shoulder which supports the thumb-under technique. Our previous user studies [Nijima et al. 2022, 2021] have shown that our system aids novices in learning to play tremolos and scales efficiently, resulting in reduced muscle activity for tremolos and improved muscle synergies for scales.

2 IMPLEMENTATION

To facilitate efficient motor learning for novices, our implementation focuses on two primary challenges: reducing forearm muscle activity when playing tremolos and improving muscle synergies when playing scales. The novelty of this system is the use of EMS to move the wrist and shoulder instead of the fingers to make novices aware of the muscle coordination of the fingers and arms.

2.1 Hardware

Our system consists of an electric piano, a PC, an electrical stimulator, and a MIDI controller (Figure 1). Our previous studies used a function generator and isolators, but they were not portable, so we developed an original compact 4-channel electrical stimulator. The electrical stimulator includes a microcomputer (Arduino) so that the parameters of EMS such as pulse width, frequency, and

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Figure 1: Setup of proposed EMS-based system.

duration can be controlled by the MIDI controller. The stimulator also uses the MIDI data of the electric piano as the trigger for EMS.

The specifications of the electrical stimulator are as follows. The size is 199 mm × 202 mm × 74 mm. A portable battery is used for the power supply. The electrical stimulation is a constant voltage at 0–60 V. The pulse width range is 16 μs–1 ms, and the frequency range is 1 Hz–1 kHz. The default parameters are 200 μs and 200 Hz, respectively. For safety, the stimulator includes a reset IC and a physical button to immediately turn off EMS.

2.2 Playing tremolos with EMS

Our strategy is to target the supinator and pronator teres muscles, which are crucial for wrist rotation (Figure 2), a technique often used by expert pianists when performing tremolos. We induced involuntary wrist rotation by alternately applying EMS to these muscles. This intervention enables novices to become familiar with wrist rotation while playing tremolos. Over time, they can adopt this efficient technique, resulting in less reliance on finger movements and consequently avoiding muscle fatigue.

2.3 Playing scales with EMS

The second component of our system is aimed at improving muscle synergy when playing the C major scale on the piano, with particular emphasis on the smooth execution of the thumb-under technique when transitioning between the E and F keys, i.e., passing the thumb under the palm before pressing the F key with the thumb (Figure 2). The smoothness of note playing depends largely on the coordination of movements, such as pressing each key and sliding the hand sideways at the same time. To facilitate this coordination, our EMS-based system is designed to apply EMS (pulse width: 200 μs, frequency: 200 Hz, duration: 200 ms) to the deltoid muscle to support the simultaneous slight rotation of the elbow when pressing the E key, thus enabling the smooth execution of the thumb-under technique before pressing the F key.

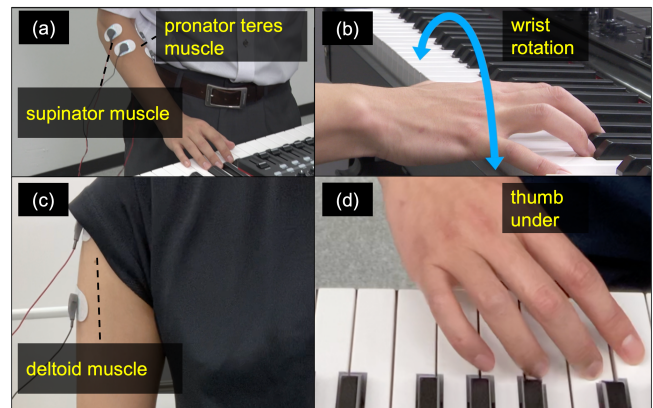


Figure 2: EMS assistance for playing tremolos and scales. (a) Electrodes placed on the supinator and pronator teres muscles. (b) Wrist rotation induced by EMS for playing tremolos. (c) Electrodes placed on the deltoid muscle. (d) EMS on deltoid muscle assisting with thumb-under technique for playing scales.

2.4 Demonstration

First, we will demonstrate our EMS-based system for playing tremolos and scales by applying EMS to ourselves. Participants will be able to control the parameters of EMS with the MIDI controller and observe how EMS controls their muscles. Next, if the participants wish to try the system, we will ask them to follow our safety instructions, carefully calibrate the voltage of the EMS to avoid pain, and sign a consent form before taking part.

3 CONCLUSION

We have presented two EMS-based systems to assist novice piano learners with playing tremolos and scales. Our first system targets the supinator and pronator teres muscles using EMS to guide wrist rotation while playing tremolos. Our second system aims to improve the smoothness of scale playing by focusing on the coordination of the forearm and shoulder muscles. This system applies EMS to the deltoid muscle to help novices perform the thumb-under technique smoothly. We believe that our system will lead to a new method of motor-skill transfer by focusing on muscle coordination.

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